

Math 340: Elementary Matrix and Linear Algebra

PRACTICE MIDTERM TWO

Name: _____ Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a 90-minute exam. It consists of five problems on six pages; the exam is four sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculator and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process.
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (4+4+5 points) Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}$$

be four vectors from $\mathbb{R}^3$.

(a) Let $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a set of vectors. Is S_1 linearly independent?

(b) Let $S_2 = \{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ be another set of vectors. What is the dimension of $\text{span} S_2$?

(c) Let $V = \text{span} S$, where

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \right\}.$$

Find a basis for V as a subspace of $\mathbb{R}^{2 \times 2}$.

2. (4+5 points)

- (a) Let $B = (t - 1, t^2 - 1, 2t + 2)$ be an ordered basis for P_2 and $p = 3t^2 - 7t - 12$ be a polynomial in P_2 . Find $[p]_B$.

- (b) Let $f: \mathbb{R}^3 \rightarrow P_1$ be a function defined by

$$f\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = (a - b)t + (c + a).$$

- i. Is f injective (one-to-one)?
- ii. Is f surjective (onto)?
- iii. Is f an isomorphism?

3. (5+5 points) Let

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -3 & 1 \\ 1 & -6 & -7 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & 2 \\ (c^2 - 2) & 0 & 4 \\ 2 & 1 & 3 \end{bmatrix}$$

be two 3×3 matrices.

(a) Determine the rank and the nullity of A .

(b) Find c if the dimension of the row space of C is 2.

4. (5+5 points)

(a) Let D be a 5×4 matrix whose rank is 3. Is it possible that

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \end{bmatrix} \text{ and } \mathbf{x}_2 = \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

are both solutions of $D\mathbf{x} = \mathbf{0}$? State yes or no, and justify your answer.

(b) Let W be a subspace of P_3 , where

$$W = \{at^3 + bt^2 - at + b \mid a, b \in \mathbb{R}\}.$$

- i. Find a basis for W .
- ii. What is $\dim W$?

5. (4+4 points) Make sure to justify your answer to the following questions.

(a) Let $L: P_2 \rightarrow P_3$ be a function defined by

$$L(at^2 + bt + c) = (a + c)t^3 - b + 1.$$

Is L a linear transformation?

(b) Let $K: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a function defined by

$$K\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} 2u_2 - u_3 \\ u_3 + 3u_1 \end{bmatrix}$$

Is K a linear transformation?

6. (5+5 points)

(a) Let W be the following subset of $\mathbb{R}^3$:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid xz = 1 \right\}.$$

Is W a subspace of $\mathbb{R}^3$ with the usual vector addition and scalar multiplication?

(b) Let P_* be a subset of P defined as follows:

$$P_* = \{ a_2 t^2 + a_1 t + a_0 \mid a_0, a_1, a_2 \in \mathbb{R} \text{ and } a_0 + a_2 = 0 \}.$$

Is P_* a vector space with the usual polynomial addition and scalar multiplication?